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标题	Metal – N-Heterocyclic Carbene Complexes in Buchwald – Hartwig Amination Reactions
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摘要（原文节选）

We present a comprehensive overview of the Buchwald – Hartwig amination, one of the most useful methods for C – N bond formation, mediated by NHC – transition-metal-complexes, covering the literature since 1999 (the first report on Buchwald – Hartwig amination by Nolan et al.) through December 2024. Palladium – and nickel – N-heterocyclic carbene (NHC) complexes are key contributors to Buchwald – Hartwig amination and are thoroughly discussed in this review, along with examples of cobalt and rhodium – NHC complexes. Apart from th ⋯

数据表 · 1（7 行，含数值行 7 行）

Comparison of Pd/Phosphine and Pd/NHC Ligand Systems by Trudell

entry	Pd/ligand	time (h)	yield (%)
1	Pd(OAc)2/PPh3	42	37
2	Pd(OAc)2/dppp	40	25
3	Pd(OAc)2/dppb	40	11
4	Pd2(dba)3/P(o-Tol)3	36	51
5	Pd2(dba)3/dppf	36	66
6	Pd2(dba)3/DiIMes · HCl	36	61
7	Pd2(dba)3/DiIPr · HCl	36	67

脚注（逐字）：adppp = 1,3-bis(diphenylphosphino)propane, dppb = 1,4-bis(diphenylphosphino)butane, dppf = 1,1′-bis(diphenylphosphino)ferrocene.

正文对应句（逐字）：Subsequently, the same authors took the advantage of these monocarbenepalladium(0) complexes, [Pd(IMes)(dvds)], [Pd(IMes)(BQ)]2, and [Pd(IMes)(NQ)]2, in the Buchwald – Hartwig cross-coupling of 4-chloro- and 4-bromoanisole with mesitylamine (Scheme). Among the three catalysts tested, [Pd-(IMes)

数据表 · 2（14 行，含数值行 14 行）

Comparison between Different NHC – Pd Complexes in BHA reaction by Nechaev

entry	NHC	yield (%)
1	[Pd(PPh3)4]	-
2	[Pd(PPh3)2Cl2]	-
3	[Pd(P(o-Tol)3)2Cl2]	-
4	Pd(dba)2+SPhos	-
5	Pd(OAc)2+SPhos	-
6	Pd(OAc)2+RuPhos	-
7	[Pd(SIPr)(cin)Cl]	-
8	[Pd(IPr)(cin)Cl]	-
9	[Pd(IPr)(PEPPSI)]	-
10	[Pd(THP-Mes)(cin)Cl]	95
11	[Pd(THP-Dipp)(cin)Cl]	>99
12	[Pd(THD-Mes)(cin)Cl]	93
13	[Pd(THD-Dipp)(cin)Cl]	-
14	Pd(OAc)2 + IPr · HCl	-

脚注（逐字）：aTraces of diphenylamine observed.

正文对应句（逐字）：Subsequently, the same authors took the advantage of these monocarbenepalladium(0) complexes, [Pd(IMes)(dvds)], [Pd(IMes)(BQ)]2, and [Pd(IMes)(NQ)]2, in the Buchwald – Hartwig cross-coupling of 4-chloro- and 4-bromoanisole with mesitylamine (Scheme). Among the three catalysts tested, [Pd-(IMes)

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